

CS 210: Data Management for Data Science

Course Project Framework

Grading Weight: 10% of Total Course Grade

Grade Distribution Overview

Component	Weight	Points (out of 100)
Project Proposal + Project Oral Exam	25%	25
Final Report + Demo Video & Implementation	75%	75
Total	100%	100

1 Part 1: Project Proposal + Project Oral Exam

Weight: 25 points

Project Proposal Due: February 25th, 2026, 11:59 PM

Project Oral Exam Due: April 26th, 2026, 11:59 PM

1.1 Grading : Please refer to table in the end

2 Part 2: Final Report, Demo Video & Implementation

Weight: 75 points

Due: TBA

2.1 Grading Rubric : Please refer to table in the end

2.1.1 A. Written Report (45 points)

2.1.2 B. Technical Implementation (25 points)

2.1.3 C. Project Demo Video, Github Repo & Innovation (5 points)

3 Additional Policies

3.1 Late Submission Policy

- Deadlines are final. No late submissions accepted.
- Medical/emergency extensions require documentation

3.2 Academic Integrity

- AI tools can assist but not generate content directly
- All sources, tools, and papers must be cited
- Plagiarism results in 0 points and academic misconduct referral
- Code must be original (libraries/frameworks excepted)

3.3 Collaboration Guidelines

- Groups of 1-3 students (3 recommended)
- All group members receive same grade unless peer evaluation indicates otherwise
- Individual contributions must be documented in final report

3.4 Submission Requirements

- **Proposal:** PDF via codebench
- **Final Report:** PDF + Demo video + code repository (GitHub/Codebench)
- Code submission via codebench must include: All code files, data files, README, requirements.txt, example data/instructions

4 Example Grading Scenarios

4.1 Scenario 1: Excellent Project (95/100)

- Proposal: 24/25 (minor citation issue)
- Report: 44/45 (comprehensive, well-written)
- Implementation: 24/25 (excellent code, strong performance)
- Innovation: 5/5 (creative approach)

4.2 Scenario 2: Good Project (80/100)

- Proposal: 20/25 (good but basic literature review)
- Report: 37/45 (solid analysis, some gaps in discussion)
- Implementation: 19/25 (works well, some code quality issues)
- Innovation: 4/5 (standard approach, well-executed)

4.3 Scenario 3: Satisfactory Project (65/100)

- Proposal: 16/25 (vague methodology)
- Report: 30/45 (basic analysis, limited insights)
- Implementation: 16/25 (works but errors, basic evaluation)
- Innovation: 3/5 (minimal creativity)

5 Success Tips

1. **Start Early:** Begin exploring datasets and questions by mid-February
2. **Get Feedback:** Discuss proposal with instructor before submission
3. **Iterate:** Use proposal feedback to refine final approach
4. **Document:** Keep detailed notes on decisions and experiments
5. **Validate:** Test code thoroughly before submission
6. **Visualize:** Invest time in creating clear, informative plots
7. **Reflect:** Honest discussion of limitations shows maturity

Questions? Contact course staff during office hours or via email.

Criteria	Excellent (4-5)	Good (3-3.5)	Satisfactory (2-2.5)	Needs Improvement (0-1.5)	Points
Problem Definition & Relevance (5 pts)	Clear, well-defined problem with explicit connection to course concepts. Strategic aspects well-articulated.	Problem defined with some connection to course material. Strategic aspects mentioned.	Problem somewhat vague. Weak connection to course material.	Problem unclear or no connection to course content.	/5
Novelty & Importance (5 pts)	Compelling justification of importance. Thorough literature review. Clear gap identification.	Good justification with some prior work mentioned.	Basic justification. Minimal literature review.	Weak justification or no prior work discussed.	/5
Data Description (5 pts)	Detailed data source, format, features clearly specified. Data accessibility confirmed.	Data source and format identified. Most features described.	Vague data description. Uncertain accessibility.	Data source unclear or infeasible.	/5
Methodology & Plan (7 pts)	Detailed implementation plan with clear evaluation metrics. Appropriate method selection with justification.	Clear plan with evaluation approach. Method selection reasonable.	Basic plan with limited detail on evaluation.	Vague or inappropriate methodology.	/7
Writing Quality & Structure (3 pts)	Excellent organization, clear writing, proper citations, within page limit.	Good organization, minor issues, proper citations.	Acceptable writing with some organizational issues.	Poor organization, unclear writing, missing citations.	/3
Total Proposal Points:					/25

Criteria	Excellent (9-10)	Good (7-8)	Satisfactory (5-6)	Needs Improvement (0-4)	Points
Problem Definition & Background (10 pts)	Refined, comprehensive problem statement with strong course connections. Excellent literature review.	Clear problem definition with good background. Adequate literature review.	Basic problem definition. Limited background.	Unclear problem or weak background.	/10
Data & Methodology (10 pts)	Comprehensive data description. Method selection well-justified with clear rationale for choices.	Good data description. Reasonable method justification.	Basic data description. Limited method justification.	Insufficient data description or inappropriate methods.	/10
Results & Analysis (15 pts)	Insightful analysis with clear interpretation. Results well-presented with effective visualizations. Strong connection between findings and hypothesis.	Good analysis with reasonable interpretation. Clear results presentation.	Basic analysis. Results presented but limited interpretation.	Weak analysis or unclear results.	/15
Discussion & Limitations (7 pts)	Thorough discussion of advantages, limitations, and future improvements. Critical self-reflection.	Good discussion of key limitations and some improvements suggested.	Basic discussion of limitations.	Minimal or no discussion of limitations.	/7
Writing & Presentation (3 pts)	Excellent organization, clarity, proper citations, appropriate length.	Good quality with minor issues.	Acceptable quality with some problems.	Poor quality or major issues.	/3

Criteria	Excellent (5)	Good (3.5-4)	Satisfactory (2-3)	Needs Improvement (0-1.5)	Points
Code Quality & Documentation (5 pts)	Well-structured, commented code. Clear README. Reproducible.	Good structure with adequate documentation.	Basic documentation. Some reproducibility issues.	Poor code quality or not reproducible.	/5
Correct Implementation (10 pts)	Flawless implementation of techniques. Best practices followed.	Correct implementation with minor issues.	Implementation works but has notable issues.	Major implementation errors.	/10
Model Performance / Insights (10 pts)	Excellent performance OR deep insights (unsupervised). Comprehensive evaluation.	Good performance/insights with proper evaluation.	Acceptable performance/insights with basic evaluation.	Poor performance/insights or inadequate evaluation.	/10

Criteria	Excellent (5)	Good (3.5-4)	Satisfactory (2-3)	Needs Improvement (0-1.5)	Points
Visualization Quality (2.5 pts)	Outstanding visualizations that clearly communicate insights. Professional quality.	Good visualizations that support findings.	Basic visualizations present.	Poor or missing visualizations.	/2.5
Creativity & Innovation (2.5 pts)	Novel approach or creative solution. Goes beyond basic implementation.	Some creative elements or thoughtful approach.	Standard approach with minimal creativity.	No innovation; purely mechanical implementation.	/2.5
Total Final Report Points:					/75